

# Arnav Pundir

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## PROFESSIONAL SUMMARY

Full-Stack Software Engineer proficient in building high-performance, real-time, and offline-first web systems. Experienced in React 19, Next.js 15, Node.js, Python (Flask), TypeScript, RxDB, and WebRTC. Strong expertise in engineering low-latency architectures, 3D WebGL performance, vector embeddings, state synchronization, and delivering production-grade client applications.

## SKILLS

**Languages:** TypeScript, JavaScript, Python, C++, SQL, HTML5/CSS3  
**Frontend & WebGL:** React 19, Next.js 15, Three.js (WebGL), GSAP, Tailwind CSS, RxDB (Offline PWA), Framer Motion  
**Backend & AI Systems:** Node.js, Express.js, Flask, FastAPI, WebRTC, Redis, WebSockets, InsightFace (ArcFace), Gemini AI  
**Databases & Infrastructure:** Supabase (PostgreSQL / pgvector), IndexedDB, MongoDB, Clerk Edge Auth, Technical SEO & Web Vitals, Docker, Git, Vercel

## EDUCATION

**COER University** Roorkee, India  
*Bachelor of Technology in Computer Science and Engineering* 2024 – 2028

## PROFESSIONAL EXPERIENCE

**Freelance Full-Stack Developer** Jan 2024 – Present  
*SparkTech Web Solutions (Client & Contract Systems)* Roorkee, India

- Engineered commercial web platforms (e.g., SevaSetu) using **React.js** and REST APIs, serving **1,000+ active end-users**.
- Architected interactive 3D WebGL websites utilizing Three.js and GSAP, achieving **95+ Lighthouse performance scores**.
- Implemented route-level lazy loading and JSON-LD Technical SEO, cutting initial asset payload by **40%** and accelerating FCP.

## PROJECTS

**ZeroLag Offline-First Collaborative Platform** [Live](#) | [GitHub](#)

- Architected a local-first task PWA using **React 19**, TypeScript, and RxDB (IndexedDB), achieving **zero perceived UI latency** via optimistic state updates.
- Built an event-sourced background sync engine with **Supabase PostgreSQL** and Realtime WebSockets for live presence updates.
- Integrated **Gemini Vision AI** via Vercel proxy to parse whiteboard photos into tasks, secured with Clerk JWT + PostgreSQL RLS.

**BioSecure AI Full-Stack Biometric Platform** [GitHub](#)

- Deployed **InsightFace ArcFace 512D** face embeddings paired with Supabase pgvector search in PostgreSQL at **90%+ accuracy** on live video feeds.
- Engineered a 3D Pose-Gated EWMA Embedding Drift Engine ( $\alpha = 0.30$ ) to monitor template aging and mitigate biometric degradation.
- Selected for official **COER University Pilot Run**; built a production Flask app with multi-role access control, automated SMTP alerts, and CSV exports.

**AuraMeet Real-Time Video Platform** [Live](#) | [GitHub](#)

- Architected a peer-to-peer video streaming app using **Node.js**, WebRTC, and Redis delivering **sub-100ms low-latency** communication.
- Designed an ephemeral data lifecycle using **Redis pub/sub** that automatically purges chat logs and room state upon disconnect.
- Programmed a dynamic video canvas grid to optimize screen rendering performance across desktop and mobile devices.

## LEADERSHIP & OPEN SOURCE CONTRIBUTIONS

**Competitive Coding Committee Lead** Jan 2025 – Present  
*D-Coders Squad (Official Tech Club of COER University)* Roorkee, India

- Spearhead algorithmic problem-solving workshops and mentor **100+ students** in core Data Structures & Algorithms (DSA).
- Organize inter-departmental coding battles and co-led executive pilot sensitization sessions for BioSecure AI in front of university leadership.

**Core Contributor** Apr 2026 – Present  
*Nexus Spring of Code (NSoC '26) & GSSoC* Global Rank #70

- Achieved **Global Rank #70** by merging 13 Pull Requests across 5 open-source repositories, improving security and performance.
- Migrated legacy authentication handlers to secure, rotating **HttpOnly cookie sessions** and tightened CORS security policies.